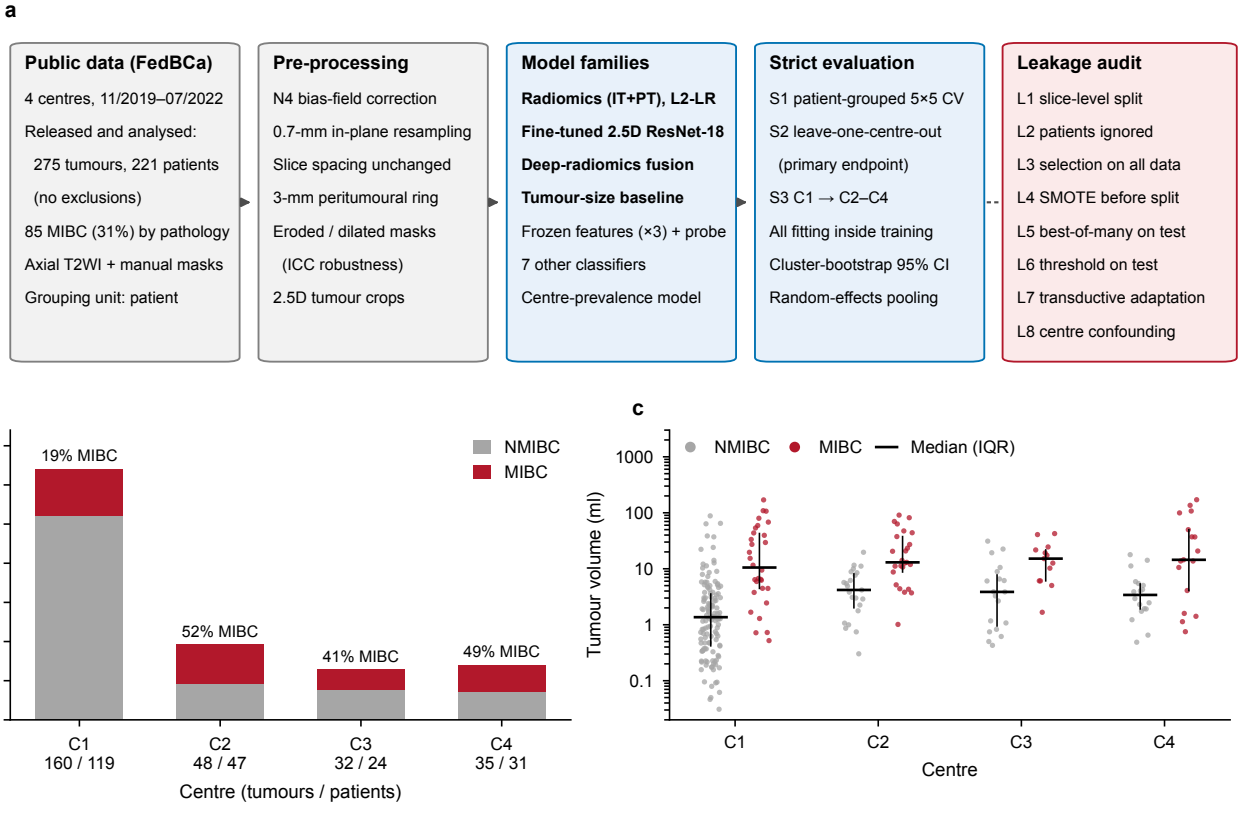

a
Public data (FedBCa)
4 centres, 11/2019–07/2022
Released and analysed:
275 tumours, 221 patients
(no exclusions)
85 MIBC (31%) by pathology
Axial T2WI + manual masks
Grouping unit: patient
Pre-processing
N4 bias-field correction
0.7-mm in-plane resampling
Slice spacing unchanged
3-mm peritumoural ring
Eroded / dilated masks
(ICC robustness)
2.5D tumour crops
Model families
Radiomics (IT+PT), L2-LR
Fine-tuned 2.5D ResNet-18
Deep-radiomics fusion
Tumour-size baseline
Frozen features (×3) + probe
7 other classifiers
Centre-prevalence model
Strict evaluation
S1 patient-grouped 5×5 CV
S2 leave-one-centre-out
(primary endpoint)
S3 C1 → C2–C4
All fitting inside training
Cluster-bootstrap 95% CI
Random-effects pooling
Leakage audit
L1 slice-level split
L2 patients ignored
L3 selection on all data
L4 SMOTE before split
L5 best-of-many on test
L6 threshold on test
L7 transductive adaptation
L8 centre confounding
b
Tumours (n)
NMIBC
MIBC
19% MIBC
52% MIBC
41% MIBC
49% MIBC
C1
160 / 119
C2
48 / 47
C3
32 / 24
C4
35 / 31
Centre (tumours / patients)
c
NMIBC
MIBC
Median (IQR)
Tumour volume (ml)
C1
C2
C3
C4
Centre